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Fiscal rules as a response to commodity shocks: A welfare analysis of the Colombian scenario☆

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ABSTRACT

We analyze the welfare effects of alternative fiscal rules in the context of shocks to the commodity sector of a commodity-rich country. We build a DSGE model featuring three productive sectors (non-tradable, manufacturing and commodities), government and two types of consumers, according to whether or not they have access to financial markets (Ricardians and non-Ricardians, respectively). The model is calibrated and estimated using Bayesian methods and macroeconomic data from Colombia. Our results show that while non-Ricardian consumers have welfare gains from countercyclical fiscal rules, Ricardian consumers incur in considerable welfare losses from this type of policy. The specific quantitative results of our welfare evaluation are driven by the structural parameters estimated for the Colombian economy. In particular, the estimated persistence of the productivity shock is relatively low and implies lower welfare changes compared to similar studies for other countries.

1. Introduction

Recently, Norway, Chile and Colombia, among other countries, formally incorporated fiscal rules within their economic policy tools. There are at least three reasons that justify implementing this kind of policy: lack of fiscal discipline, low credibility and volatility (García et al. (2011)). The first reason is related to the presence of weak institutions and controls which are supposed to prevent excessive fiscal deficits. A fiscal rule may contribute to improve the expectations of fiscal discipline by market participants and thus increase the credibility of fiscal commitments. The third reason deals with the abrupt adjustments of revenues and expenditures that are needed when commodity shocks hit in the absence of fiscal rules. While governments have implemented fiscal rules obeying these considerations, it is necessary to perform welfare evaluations of their overall economic effect. This is precisely the goal of our study.

Individuals and firms, especially in high income-volatility countries, try to sustain their levels of consumption and investment through their access to financial markets, by selling assets or borrowing during economic busts, or by accumulating assets or repaying debt during booms. However, there may be a significant proportion of people, particularly in emerging countries, who do not have such access, resulting in economic inefficiencies. This is one of the reasons why a countercyclical fiscal policy (i.e. saving in good times and spending in bad times), by partially compensating for the lack of access to credit markets, can be a useful tool for the optimal management of commodity-linked income.

In order to contribute to the study of economic cycles and fiscal policy in a small open commodity-dependent country, we analyze the commodity sector and its interaction with the rest of the economy in a dynamic stochastic general equilibrium (DSGE) framework with three private sectors (a non-tradable sector and two tradable ones, manufacturing and commodity) plus a government sector and two types of consumers, those with access to the financial market (Ricardian) and those without such access (non-Ricardian).

We analyze how macroeconomic variables react to a commodity productivity shock when different degrees of cyclicality of a fiscal policy rule are considered. In particular, we study rules with different degrees of pro- or countercyclicality and their particular effects on social welfare. For this purpose, we estimate some of the model's parameters by Bayesian methods using quarterly data of the main Colombian[1] macroeconomic variables. The remaining parameters are fixed or calibrated in

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[1] Colombia is a recent example of a commodity-exporting country experiencing a boom in this sector. On the one hand, in 2011 more than 60% of Colombian exports corresponded to the mining-energy sector (mainly oil and coal). On the other hand, the average crude oil production in Colombia in March 2012 was approximately 951,000 barrels per day, that is, 7.2% above the production in the same month of 2011, which in turn was 13% higher than the average production in 2010.

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line with stylized facts of the Colombian economy and based on the related literature.

We find that non-Ricardian consumers are better off under a countercyclical fiscal rule. Since non-Ricardian consumers cannot use the financial market to intertemporally smooth consumption, they benefit from the management of income made by the government through countercyclical transfers. On the other hand, Ricardian consumers incur in considerable welfare losses from this type of policy. The specific quantitative results of our welfare evaluation are driven by the structural parameters estimated for the Colombian economy. In particular, the estimated persistence of the productivity shock is relatively low and implies lower welfare gains than similar studies for other countries.

The following section presents a literature review on fiscal policy responses to commodity shocks. The third section describes the model and its equilibrium conditions. The fourth corresponds to the empirical estimation of the model parameters. The fifth section shows the analysis of the impulse-response functions of some macroeconomic variables under alternative fiscal rules. The welfare analysis is presented in the sixth section. The seventh section concludes.

2. Literature review

Commodity-rich countries may tend to experience high economic volatility and, as a consequence, policy interventions may be necessary to moderate cycles. In this regard, fiscal policy is seen as an important instrument for stabilization.

Policymakers from countries experiencing commodity shocks have faced serious concerns because of its potential effects on macroeconomic stability, the competitiveness of the export sector, and the external viability of the recipient countries. Institutional arrangements cannot be excluded from the analysis, as evidence shows that social norms, a social contract, transparency and rule of law may contribute to limiting rent seeking (Røed-Larsen, 2006). One of the pioneering works on studying the macroeconomic effects of a booming sector is Corden and Neary (1982) who are interested in the medium-run effects of asymmetric growth on resource allocation and income distribution.

The models that look at the role that fiscal policy could play in managing that income are of particular interest to the present paper. Most of them seek to answer the question of what is the appropriate framework for fiscal policy in order to mitigate the potential costs of commodity price fluctuations, where “appropriateness” is determined by its welfare effects. By analyzing data for Mexico and Norway, Pieschacón (2012) shows that fiscal policy is a key transmission channel that affects the degree of exposure to oil shocks. She finds that fiscal policies that insulate the country from exogenous oil price shocks seem to be welfare improving over those that are procyclical. As some previous literature has remarked (e.g. Frankel et al. (2013), Céspedes and Velasco (2014)), while the developed world has pursued countercyclical or acyclical fiscal policies, developing countries and particularly commodity-rich countries have tended to be procyclical. However, over the last decade, the improvement in the quality of institutions and the presence of fiscal rules in some of these developing countries have reduced the procyclicality of their fiscal policies.

The Chilean economy has been an interesting case for this literature, since an important source of its income comes from copper production and exports, and the country has implemented a fiscal rule as part of the mechanisms to manage this income.[2] Various DSGE models have compared the effect of transitory copper-price shocks under different fiscal rules, for example a balanced-budget rule and a structural rule (see, among others, Medina and Soto (2007) and García and Restrepo (2007)). In a model with New Keynesian features and with some non-Ricardian (hand-to-mouth) households, García et al. (2011) find that an acyclical fiscal rule benefits households that do not enjoy access to capital markets but reduces welfare for households that enjoy full access to capital markets. A fiscal policy in between a procyclical balanced-budget policy and an acyclical structural surplus may be preferred by all agents (see also Kumhof and Laxton (2010)).

In this paper, we propose a dynamic stochastic general equilibrium model based on Acosta et al. (2009) but including explicitly a productive commodity sector that uses capital and labor in its production process.[3] We follow Corden and Neary (1982) in assuming sector-specific capital accumulation. Furthermore, based on García et al. (2011), we include a fiscal sector with different reaction rules to the business cycle. We evaluate the effect of these rules on social welfare.

The major contribution of this paper to the literature is to make a quantitative analysis of the commodity shocks and fiscal policy in the case of Colombia. With that aim, we explicitly model the commodity production, taking into account its input utilization, making the analysis of general equilibrium more realistic. Additionally, most of the model's parameters are estimated with Bayesian methods (using macroeconomic data for Colombia), and we perform a welfare analysis of alternative fiscal rules in the context of commodity productivity shocks.

3. The model

The model consists of three types of agents: households, firms and government. The problem faced by each type is described in detail in this section. The model structure is based on Acosta et al. (2009), who propose a model with two sectors (non-tradable and tradable or manufacturing) to analyze the impact of remittances on a small open economy. We extend their model by including the commodity sector and the government through the fiscal policy and by differentiating households according to their access to the financial market.

3.1. Households

There is a continuum (of measure one) of identical households that, every period, decide their level of real consumption and labor supply in a perfectly competitive market. There are two types: Ricardian, who are a proportion z of households and non-Ricardian, who are the rest, $1–z$. Ricardian agents (R) have access to the financial market and therefore they also decide about the level of debt/savings (bonds) and their participation in firms (shares) for the next period. In contrast, non-Ricardian (NR) cannot save or borrow and their income is completely determined by their participation in the labor market. There is no other difference between Ricardian and non-Ricardian households in the model, apart from their access to financial markets.

The household's utility in period t is represented by Cobb–Douglas preferences on the consumption index C_t and work effort L_t:

$$u\left(C_t^h, L_t^h\right) = \frac{\left[\left(C_t^h\right)^{(1-\omega)}\left(1-L_t^h\right)^{\omega}\right]^{1-\gamma}-1}{1-\gamma}$$

where $h = R, NR$, $\omega \in [0, 1]$ and $\gamma > 0$. The index of consumption C is a composite of non-tradable (C_N^h) and tradable (C_T^h) consumption. In turn, tradable consumption is a composite of the consumption of

[2] Frankel (2011) examines the problem of procyclicality with respect to both fiscal and monetary policy in countries that depend on primary commodities. With regard to fiscal policy, he proposes to emulate Chile's structural rule as a way to escape from procyclicality.

[3] In a multi-country model Backus and Crucini (2000) endogenize the commodity (oil) sector which uses only labor as input. They study the relation between oil prices and terms of trade. There are neither non-tradable goods nor a fiscal sector in their model. Similarly to our model, their model abstracts from the use of oil by the oil-producing country.

domestic manufactured goods (C_M^h) and imported goods (C_F^h).[4] We assume that the domestic production of commodities is completely exported and, therefore, its domestic consumption is zero.

Households maximize:

$$E_0\left\{\sum_{t=0}^{\infty} exp\left[\sum_{\tau=0}^{t-1}\beta_\tau\right] u\left(C_t^h,\ L_t^h\right)\right\} \tag{1}$$

where $\beta_t = -\kappa \log(1 + C_t^{(1-\omega)}(1-L_t)^{\omega})$ and, therefore, the intertemporal discount factor is endogenously determined.[5] It ensures that the model has a stable solution, as shown by Schmitt-Grohé and Uribe (2003). The endogenous discount factor has been used in previous works such as Mendoza (1991) and Acosta et al. (2009).

The budget constraint for the Ricardian households is:

$$\begin{aligned} P_t C_t^R + \upsilon_{M,t}\chi_{M,t+1} + \upsilon_{E,t}\ \chi_{E,t+1} \leq & (\upsilon_{M,t} + d_{M,t})\chi_{M,t} \\ & +(\upsilon_{E,t} + d_{E,t})\chi_{E,t} + w_t L_t^R - B_t + \frac{1}{1+r}B_{t+1} + Tr_t \end{aligned} \tag{2}$$

where P_t is the consumer price index (which is a composite of the price of non-tradable goods $P_{N,t}$ and the price of tradable goods $P_{CT,t}$).[6] $\chi_{M,t+1}$ and $\chi_{E,t+1}$ are the household's shares, purchased in t, in the firms' profits in manufacturing and commodity sectors, respectively. The unit prices for such shares are $\upsilon_{M,t}$ and $\upsilon_{E,t}$ and their respective dividends are $d_{M,t}$ and $d_{E,t}$. w_t represents the wage, expressed in units of the manufactured good. International financial transactions are made through one-period risk-free bonds (B_t). The international interest rate is constant and equal to r. Tr_t corresponds to government lump-sum transfers to consumers (see Section 3.3).

The solution to the Ricardian consumer's problem can be obtained from the following first order conditions with respect to bonds B_{t+1}, shares $\chi_{M,t+1}$ and $\chi_{E,t+1}$, labor supply L_t and consumption, respectively:

$$1 = exp(\beta_t)E_t\left[\frac{\lambda_{C,t+1}}{\lambda_{C,t}}(1+r)\frac{P_t}{P_{t+1}}\right] \tag{3}$$

$$\upsilon_{M,t} = exp(\beta_t)E_t\left[\frac{\lambda_{C,t+1}}{\lambda_{C,t}}(\upsilon_{M,t+1} + d_{M,t+1})\frac{P_t}{P_{t+1}}\right] \tag{4}$$

$$\upsilon_{E,t} = exp(\beta_t)E_t\left[\frac{\lambda_{C,t+1}}{\lambda_{C,t}}(\upsilon_{E,t+1} + d_{E,t+1})\frac{P_t}{P_{t+1}}\right] \tag{5}$$

$$\frac{\omega}{1-\omega}\frac{C_t^R}{\left(1-L_t^R\right)} = \frac{w_t}{P_t} \tag{6}$$

$$\frac{C_{N,t}^R}{C_{T,t}^R} = \frac{\gamma_c}{1-\gamma_c}\left(\frac{P_{N,t}}{P_{CT,t}}\right)^{-\rho_c}, \frac{C_{M,t}^R}{C_{F,t}^R} = \frac{\gamma_m}{1-\gamma_m}\left(\frac{1}{P_{F,t}}\right)^{-\rho_m} \tag{7}$$

where the marginal utility of consumption is $\lambda_{C,t} = (1-\omega)(C_t^R)^{(\gamma-1)(\omega-1)-1}(1-L_t^R)^{\omega(1-\gamma)}$. $P_{F,t}$ is the domestic price[7] of imported goods (consumption or investment). The price of the domestic manufactured good is used as numeraire.

Since non-Ricardian consumers do not have access to the financial market, their budget constraint is given by $P_t C_t^{NR} = w_t L_t^{NR} + Tr_t$. As a result, the first order conditions of their problem are equivalent to Eqs. (6) and (7) evaluated at the corresponding levels of non-Ricardian consumption and labor.

3.2. Firms

The model has three sectors (non-tradable N, tradable manufacturing M, and tradable commodity sector E). Firms demand inputs (labor and capital) in perfectly competitive markets. We assume that firms in the non-tradable sector only use labor for production.

The capital accumulation for sector j, $j = \{M, E\}$, takes the form $K_{j,t+1} - K_{j,t} = I_{j,t} - \delta_j K_{j,t}$ where K is capital, I the level of investment, and δ the depreciation rate. We also assume that one unit of manufacturing good can be transformed into a unit of domestic investment good without incurring any cost. The capital of sector j is subject to installation costs $\frac{\phi_j}{2}\left(\frac{I_{j,t}}{K_{j,t}} - \delta_j\right)^2 K_{j,t}$ which are proportional to the capital stock. Labor is perfectly mobile across sectors and, as a result, wages are the same for the entire economy. The total labor demand is the sum of the labor employed in each sector, $L_t = L_{N,t} + L_{M,t} + L_{E,t}$.

3.2.1. Non-tradable sector

Non-tradable firms use labor as the only input and their production function is $Y_{N,t} = \exp(a_{N,t})L_{N,t}$, where $a_{N,t} = \bar{a}_N + \varrho_n a_{N,t-1} + \varepsilon_{n,t}$ is an exogenous stochastic process related to the sector productivity and $\varepsilon_{n,t}$ is white noise. The sector has a tax rate τ_n on the value of production. The profit maximization of this sector implies the following labor demand function:

$$\frac{(1-\tau_n)Y_{N,t}}{L_{N,t}} = \frac{w_t}{P_{N,t}} \tag{8}$$

3.2.2. Tradable manufacturing sector

Manufacturing firms demand a composite I_M of a domestic investment good I_H and a foreign (imported) investment good I_{MF}.[8] Firms produce manufactured goods using a constant return to scale technology: $Y_{M,t} = \exp(a_{M,t})K_{M,t}^{\alpha_m}L_{M,t}^{1-\alpha_m}$, where $a_{M,t} = \bar{a}_M + \varrho_m a_{M,t-1} + \varepsilon_{m,t}$ is an exogenous stochastic process related to the sector productivity and $\varepsilon_{m,t}$ is white noise. The sector has a tax rate τ_m on the value of production. The firms of this sector maximize the value of dividends as defined in Eq. (9). In turn, the dividends each period are set as the difference between the value of production and costs, as it appears in Eq. (10).

$$E_t\sum_{s=t}^{\infty} \exp(\beta_s^{s-t})\left(\frac{\lambda_{C,s}}{\lambda_{C,t}}\frac{P_t}{P_s}\right)d_{M,s} \tag{9}$$

$$d_{M,s} = (1-\tau_m)Y_{M,s} - P_{IM,s}\left(I_{M,s} + \frac{\phi_m}{2}\left(\frac{I_{M,s}}{K_{M,s}} - \delta_m\right)^2 K_{M,s}\right) - w_s L_{M,s} \tag{10}$$

[4] The total consumption index is $C_t^h = \left[(\gamma_c)^{\frac{1}{\rho_c}}(C_{N,t}^h)^{\frac{\rho_c-1}{\rho_c}} + (1-\gamma_c)^{\frac{1}{\rho_c}}(C_{T,t}^h)^{\frac{\rho_c-1}{\rho_c}}\right]^{\frac{\rho_c}{\rho_c-1}}$ and the consumption of tradable goods index is $C_{T,t}^h = \left[(\gamma_m)^{\frac{1}{\rho_m}}(C_{M,t}^h)^{\frac{\rho_m-1}{\rho_m}} + (1-\gamma_m)^{\frac{1}{\rho_m}}(C_{F,t}^h)^{\frac{\rho_m-1}{\rho_m}}\right]^{\frac{\rho_m}{\rho_m-1}}$. The parameters γ_c and γ_m correspond to the share of non-tradable goods in total consumption and the participation of domestic goods in the consumption of tradable goods, respectively. The parameters ρ_c and ρ_m represent the corresponding elasticities of substitution.

[5] We assume that the discount rate depends on the average levels of consumption and work effort of the economy and, as a result, the household takes these values as given to maximize utility. These average levels are $C_t = zC_t^R + (1-z)C_t^{NR}$ and $L_t = zL_t^R + (1-z)L_t^{NR}$. Other aggregate consumption levels are similarly calculated (e.g. C_M).

[6] The consumer price index is $P_t = [(\gamma_c(P_{N,t})^{1-\rho_c} + (1-\gamma_c)(P_{CT,t})^{1-\rho_c}]^{\frac{1}{1-\rho_c}}$ and $P_{CT,t} = [\gamma_m + (1-\gamma_m)(P_{F,t})1-\rho_m]11-\rho_m$. The price of the manufactured good is used as the numeraire and $P_{F,t}$ is the domestic price of imported goods.

[7] Since the analysis of the model is made in terms of real variables, for the sake of simplicity the value of the nominal exchange rate is normalized to one.

[8] The composite investment good is $I_{M,t} = \left[(\gamma_i)^{\frac{1}{\rho_i}}(I_{H,t})^{\frac{\rho_i-1}{\rho_i}} + (1-\gamma_i)^{\frac{1}{\rho_i}}(I_{MF,t})^{\frac{\rho_i-1}{\rho_i}}\right]^{\frac{\rho_i}{\rho_i-1}}$. The parameter γ_i corresponds to the domestic investment goods share in the manufacturing sector investment. The parameter ρ_i represents the corresponding elasticity of substitution.

P_{IM} is the investment price index in the manufacturing sector.[9] The maximization of (9), taking (10) into account and subject to capital accumulation, yields the following first order conditions with respect to capital $K_{M,t+1}$, investment $I_{M,t}$, labor demand $L_{M,t}$ and the investment composition:

$$E_t \exp(\beta_{t+1}) \left(\frac{\lambda_{C,t+1}}{\lambda_{C,t}} \frac{P_t}{P_{t+1}}\right) \times \left[P_{IM,t+1}\left(\phi_m\left(\frac{I_{M,t+1}}{K_{M,t+1}}-\delta_m\right)\frac{I_{M,t+1}}{K_{M,t+1}}-\frac{\phi_m}{2}\left(\frac{I_{M,t+1}}{K_{M,t+1}}-\delta_m\right)^2\right) + \alpha_m \frac{(1-\tau_m)Y_{M,t+1}}{K_{M,t+1}} + \lambda_{IM,t+1}(1-\delta_m)\right] = \lambda_{IM,t} \quad (11)$$

$$P_{IM,t}\left(1+\phi_m\left(\frac{I_{M,t}}{K_{M,t}}-\delta_m\right)\right) = \lambda_{IM,t} \quad (12)$$

$$(1-\alpha_m)(1-\tau_m)\frac{Y_{M,t}}{L_{M,t}} = w_t \quad (13)$$

$$\frac{I_{H,t}}{I_{MF,t}} = \frac{\gamma_i}{1-\gamma_i}\left(\frac{1}{P_{F,t}}\right)^{-\rho_i} \quad (14)$$

The variable λ_{IM} represents the shadow price of a unit of capital in the manufacturing sector.

3.2.3. Tradable commodity sector

We assume that in this sector all investment is imported ($I_{E,t} = I_{EF,t}$). The commodity is produced using a constant return to scale technology: $Y_{E,t} = \exp(a_{E,t})K_{E,t}^{\alpha_e}L_{E,t}^{1-\alpha_e}$. $a_{E,t} = \bar{a}_E + \varrho_e a_{E,t-1} + \varepsilon_{e,t}$ is an exogenous stochastic process, related to productivity, where $\varepsilon_{e,t}$ is white noise. The entire domestic production of this sector is exported. The price of the commodity, measured in manufacturing goods, is noted as P_E and there is a tax rate τ_e on the value of production. The firms in this sector also maximize the present value of dividends:

$$E_t \sum_{s=t}^{\infty} \exp(\beta_s^{s-t}) \left(\frac{\lambda_{C,t+1}}{\lambda_{C,t}} \frac{P_t}{P_s}\right) d_{E,t} \quad (15)$$

$$d_{E,t} = (1-\tau_e)P_{E,s}Y_{E,s} - P_{F,s}\left(I_{E,s} + \frac{\phi_e}{2}\left(\frac{I_{E,s}}{K_{E,s}}-\delta_e\right)^2 K_{E,s}\right) - w_s L_{E,s} \quad (16)$$

Similarly to the manufacturing sector, the maximization of (15) results in the following first order conditions with respect to capital $K_{E,t+1}$, investment $I_{E,t}$ and labor demand $L_{E,t}$:

$$E_t \exp(\beta_{t+1}) \left(\frac{\lambda_{C,t+1}}{\lambda_{C,t}} \frac{P_t}{P_{t+1}}\right) \left[P_{F,t+1}\left(\phi_e\left(\frac{I_{E,t+1}}{K_{E,t+1}}-\delta_e\right)\frac{I_{E,t+1}}{K_{E,t+1}}-\frac{\phi_e}{2}\left(\frac{I_{E,t+1}}{K_{E,t+1}}-\delta_e\right)^2\right) + \alpha_e P_{E,t+1}\frac{(1-\tau_e)Y_{E,t+1}}{K_{E,t+1}} + \lambda_{IE,t+1}(1-\delta_e)\right] = \lambda_{IE,t} \quad (17)$$

$$P_{F,t}\left(1+\phi_e\left(\frac{I_{E,t}}{K_{E,t}}-\delta_e\right)\right) = \lambda_{IE,t} \quad (18)$$

$$(1-\alpha_e)(1-\tau_e)\frac{Y_{E,t}}{L_{E,t}} = \frac{w_t}{P_{E,t}} \quad (19)$$

The parameter λ_{IE} represents the shadow price of a unit of capital in the commodity sector.

3.3. Fiscal sector

The government[10] obtains revenue (T) from taxes (τ_i) on the value of production of the manufacturing, commodity and non-tradable sectors:

$$T_t = \tau_m Y_{M,t} + \tau_n P_{N,t} Y_{N,t} + \tau_e P_{E,t} Y_{E,t}$$

The government uses this revenue to make lump-sum transfers (Tr) to households without distinction between Ricardian and non-Ricardian.[11] The government's budget constraint for each period is given by:

$$B_{G,t} = B_{G,t-1}(1+r) + Tr_t - T_t \quad (21)$$

where $B_{G,t}$ is the government debt (if it is negative, then it corresponds to government credits) at the end of period t.

Following García et al. (2011), a general fiscal rule that may consider different degrees of countercyclicality can be written as:

$$Tr_t = \overline{T} - (1+r+\mu)B_{G,t-1} + \varphi(T_t - \overline{T}) + \varepsilon_{T,t} \quad (22)$$

where $\overline{T}$ denotes the government's steady state revenue; that is, it does not contain any cyclical components. μ is an adjustment factor for the debt's interest payments, which in general can be interpreted as the debt level target[12]. In turn, φ determines the level of counter- or procyclicality of the government transfers. r is the international interest rate. $\varepsilon_{T,t}$ is white noise.[13]

Then, with $\varphi = 1$ and $\mu = 0$, this rule turns into a balanced-budget constraint in which the government always transfers all of its revenue without incurring deficits or surpluses. In this case, the government acts procyclically, reducing transfers (or raising taxes) during recessions and doing the opposite during expansions. With $\varphi = 0$ and $0 < \mu < (1+r)^{-1}$, the rule compares to an acyclical regime in which transfers are completely attached to the long-term level of revenue ($\overline{T}$). In this rule, the government saves during expansions and benefits from borrowing opportunities during recessions. We can think of a continuum of intermediate rules in which $-1 < \varphi < 1$ and $0 < \mu < (1+r)^{-1}$, thus capturing different degrees of counter- or procyclicality in transfers.

For this work, we study rules with different degrees of pro- or countercyclicality and the differences that these rules can generate on social welfare.

3.4. Aggregate constraints and external sector

The resource constraint in the manufacturing sector is given by $Y_{M,t} = C_{M,t} + I_{M,t} + X_{M,t}$. The exported component of the manufacturing sector is denoted by $X_{M,t}$. That is to say, domestic production of the manufacturing good is used for private consumption, government consumption, investment and exports. Following Acosta et al. (2009), we use the following notation for these exports: $X_{M,t} = e_t{}^{\xi} Y_{F,t}$. In this specification, $\xi > 0$ is the elasticity of exports to the real exchange rate (e_t), and $Y_{F,t}$ is the

[9] $P_{IM,t} = \left[\gamma_i + (1-\gamma_i)(P_{F,t})^{1-\rho_i}\right]^{\frac{1}{1-\rho_i}}$. As was explained at the beginning of Section 3.2, the manufacturing good can be transformed at no cost into a domestic investment good and, hence, the price of the latter is the same as that of the former (equal to one).

[10] Fiscal policy is modeled so that its effect on the economy occurs directly on macroeconomic variables. Some literature has explored the indirect effect that fiscal policy has on macroeconomic variables by boosting both consumer and business confidence. See Easaw et al. (2005), Konstantinou and Tagkalakis (2011), and Ludvigson (2004).

[11] This is a simplified depiction of fiscal policy. In reality, government expenditure takes the form of wage-related or non-wage-related consumption, and each one can have different impact on the economy as a whole. For example, Konstantinou and Tagkalakis (2011) show that increases in the government wage bill (increasing wages and/or public sector employment) have a negative and sometimes significant effect on agents' confidence. Conversely, additional non-wage-related government consumption spending has a positive and significant effect on consumer sentiment.

[12] This adjustment factor allows the government to accumulate assets (for precautionary purposes) that could be used at times of recession.

[13] See Mitchell et al. (2000) for a systematic study of tax-difference rules and tax-level rules employed on different models, considering both their theoretical properties and their comparative behavior in model simulations.

aggregate output of the rest of the world. Furthermore, the exchange rate is given by the ratio of the price of the imported consumption basket to the consumer price index[14] $e_t = \frac{P_{F,t}}{P_t}$.

The non-tradable sector production is equivalent to the consumption of the same sector. In the commodity sector, total production is exported. Aggregate output in units of manufacturing good is given by $Y_t = Y_{M,t} + P_{N,t}Y_{N,t} + P_{E,t}Y_{E,t}$. The current account is given by:

$$CA_t = -r\left(B_t + B_{G,t}\right) + \mathrm{X}_{\mathrm{M,t}} - P_{F,t}\left(C_{F,t} + \mathrm{I}_{\mathrm{E,t}} + \mathrm{I}_{\mathrm{MF,t}}\right) + P_{E,t}Y_{E,t} \tag{23}$$

4. Bayesian estimation and calibration

We estimate some of the model's parameters by Bayesian methods. The popularity of these methods has been growing not only in macroeconomics, but also in other disciplines. This technique uses a general equilibrium approach that intends to solve the identification problems of reduced-form models and has the advantage of performing well in small-sample estimations.[15]

For the estimations we use quarterly Colombian data on total real GDP, real exchange rate, real mining GDP and real central government debt from 1996Q1 to 2011Q4 (64 observations per series). We do not use data from 2012Q1 because from that moment on the fiscal rule began to be applied in Colombia. The final series are expressed in logarithms and are detrended using the Hodrick–Prescott filter.

The remaining parameters are fixed or calibrated.

4.1. Calibration

We set some parameters following the estimations by Acosta et al. (2009) for El Salvador. Others are calibrated to match some stylized facts of the Colombian economy.

The international interest rate, in steady state, is fixed so that the quarterly rate satisfies $(1+\bar{r}) = 1/0.99$. That is, in annual terms the real interest rate is approximately 4%. The value of κ, in the intertemporal discount rate, is endogenously determined to be consistent with the value of $\bar{r}$. The weight of leisure in the utility function (ω) is set to 0.3.

The steady state value of employment in the manufacturing sector is normalized to $L_{M,t} = 0.2$. For the manufacturing sector we assume an annual rate of depreciation $\delta_m = 3\,\%$. For the commodity sector the depreciation rate (δ_e) is endogenously determined to guarantee that the steady state real wage is equal across sectors. The elasticity of substitution between domestic and foreign investment goods is set to $\rho_i = 0.2$.

The parameter that measures the share of capital in the Cobb–Douglas production function is set to $\alpha_m = 0.33$ for the manufacturing sector and to $\alpha_e = 0.9$ for the commodity sector. The latter is obtained from an official report about the Colombian fiscal rule (Banco de la República, Ministerio de Hacienda y Crédito Público, Departamento Nacional de Planeación, 2010).

The installation cost of capital for the manufacturing sector (ϕ_m) is set to 2.2, following Acosta et al. (2009) and that for the commodity sector (ϕ_e) is estimated (see the next subsection).

Regarding consumption, the steady state participation of non-tradable goods in total consumption is fixed at $\gamma_c = 0.5$, the elasticity of substitution between tradable and non-tradable consumption is fixed at $\rho_c = 0.4$, and the share of domestic goods in tradable consumption is fixed at $\gamma_m = 0.4$. The inverse of the intertemporal elasticity of substitution (γ) is set to 2.0.

The proportion of non-Ricardian agents in the economy (1–z) is set to 50%.[16] The levels of public and private debt (B_G, B) are assumed to be zero in steady state. The value of taxes (τ_m, τ_n, τ_e) on the production of the manufacturing, non-tradable and commodity sectors is set to 20%, which is the effective tax rate in Colombia according to the estimation by Hamann et al. (2011).

4.2. Estimation

The following fourteen parameters are estimated using Bayesian methods:

- (Consumers) The elasticity of substitution between consumption of domestic and foreign tradable goods (ρ_m).
- (Firms) The share of domestic investment in total manufacturing investment (γ_i). Installation costs of capital for the commodity sector (ϕ_e).
- (Exports) The elasticity of exports to the real exchange rate (ξ).
- (Fiscal sector) The historical level of counter- or procyclicality of fiscal policy (φ).
- (Exogenous shocks) The persistence of the exogenous stochastic processes (ϱ_e, ϱ_m, ϱ_n, ϱ_f) and the standard deviations of the error terms ($\sigma_{\varepsilon_e}, \sigma_{\varepsilon_m}, \sigma_{\varepsilon_n}, \sigma_{\varepsilon_f}, \sigma_{\varepsilon_T}$).

We choose the prior distributions by taking into account the parameters' domain. The means of these distributions are taken from those obtained by Acosta et al. (2009) and relatively large values of standard deviations are assumed so that the kurtosis of posterior distributions mainly depends on empirical data.

Table 1 presents the estimation results. We use the estimated posterior means for the model simulation and impulse-response analysis in the next section.

5. Impulse-response results

In this section, we analyze the effects of a boom (positive productivity shock) in the commodity sector on consumption and employment for alternative fiscal rules. Using the parameters estimated and calibrated, as described in the previous section, we compute the implied impulse-response functions. All figures compare these macroeconomic responses under alternative values for the cyclicality parameter of the fiscal rule. In the figures below, we consider four different scenarios: countercyclical ($\varphi = -1$), neutral ($\varphi = 0$), historical ($\varphi = 0.81$, see Table 1), and balanced budget ($\varphi = 1$).

We simulate a transitory shock on the stochastic component of the commodity sector's productivity ($a_{E,t}$). The size of this shock is of one estimated standard deviation (8.8%) and its estimated quarterly persistence is $\varrho_e = 0.79$.

5.1. Effects on employment[17]

Fig. 1 shows the effect of the boom on the total employment of Ricardian consumers. The initial effect is positive, around 0.5% (with respect to the steady state) on impact, and reaches a peak of around 1%, six quarters after the shock. This peak is slightly higher when the fiscal policy follows a balanced-budget rule. This positive effect on Ricardian's employment is associated to the higher wage levels implied by the shock.

[14] This is a real model. The consumer price index is the value of the total consumption basket in terms of the numeraire, which is the domestic manufacturing good. See footnote 6.

[15] When compared to other techniques, for example, the Generalized Method of Moments and Maximum Likelihood methods.

[16] In order to determine this value we consider the fact that the bancarization index in Colombia (available since 2006) rose from 51.1% in 2006 to 64.6% in 2011 according to the reports on financial inclusion published by "Asobancaria" (the association of banks and other financial institutions in Colombia).

[17] Our model does not feature unemployment. Gaston and Rajaguru (2013) propose a small open economy model that incorporates realistic features of labor markets. The model predicts that a sustained improvement in the terms of trade lowers unemployment.

Table 1
Bayesian estimation results[a].

	Prior			Posterior – confidence interval		
Parameter	Distribution	Mean	SD	Mean	10%	90%
ρ_m	Gamma	0.50	0.20	0.35	0.13	0.55
γ_i	Beta	0.50	0.20	0.55	0.36	0.74
ϕ_e	Gamma	2.20	0.80	2.85	2.10	3.58
ξ	Gamma	1.10	0.60	0.32	0.10	0.54
φ	Normal	1.00	0.50	0.81	0.29	1.21
ϱ_e	Beta	0.80	0.10	0.79	0.71	0.86
ϱ_m	Beta	0.80	0.10	0.88	0.78	0.99
ϱ_n	Beta	0.80	0.10	0.88	0.80	0.96
ϱ_f	Beta	0.80	0.10	0.89	0.84	0.94
σ_{ε_e}	Inv. Gamma	0.05	∞	0.09	0.08	0.11
σ_{ε_m}	Inv. Gamma	0.05	∞	0.02	0.01	0.02
σ_{ε_n}	Inv. Gamma	0.05	∞	0.03	0.03	0.04
σ_{ε_f}	Inv. Gamma	0.05	∞	0.10	0.08	0.12
σ_{ε_T}	Inv. Gamma	0.05	∞	0.03	0.03	0.04

Source: authors' calculations
[a] Results are based on 1,000,000 iterations of the Metropolis–Hastings algorithm.

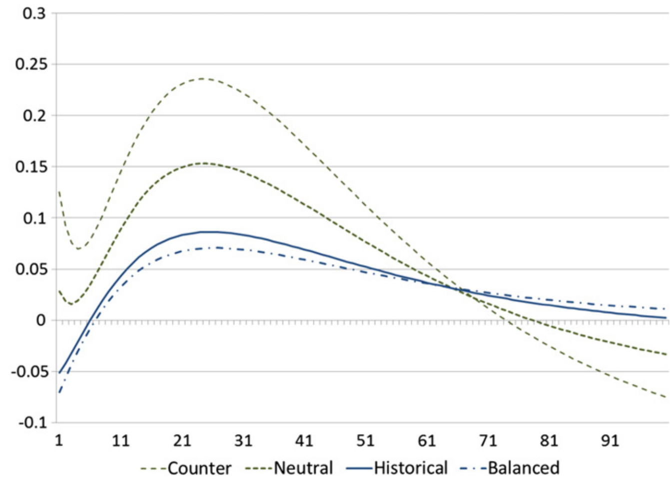

Fig. 2. Effects on non-Ricardian consumers' employment: percent deviations from steady state under each fiscal policy rule.

In Fig. 1, the positive effect on Ricardian employment lasts about 35 quarters under the balanced-budget rule and only around 25 quarters with the countercyclical fiscal rule. During the following quarters, the net effect on this employment is negative especially in the case of countercyclical rules. In a nutshell, the increase in Ricardian employment is smaller under more countercyclical rules, due to the differential effect of the shock on wages.

Fig. 2 shows the effects on the employment of non-Ricardian consumers. With a balanced-budget rule the effect is very small (always below 0.1% in absolute value). Under the countercyclical rule, there is an initial positive effect on employment lasting 60 quarters which is later compensated by a negative employment effect. The reason for this differential response of non-Ricardian labor supply (and employment) is the differential behavior of transfers received from the government under each rule. As shown in Fig. 3, under the Historical and Balanced rules, transfers received by non-Ricardian consumers increase, allowing these agents to reduce hours worked and increase consumption, leading to higher total utility. In contrast, under the countercyclical and neutral rules, government transfers initially decrease or do not increase as fast, leading non-Ricardian agents to increase their working hours to compensate for lost transfers.

Fig. 3 shows how government transfers are procyclical under a balanced-budget rule. In contrast, these transfers become initially lower if the rule is countercyclical and the shock is positive. However, once the productivity shock dissipates, government transfers remain above their steady state values during a long period of time.

5.2. Effects on consumption

Fig. 4 shows the positive effect of the commodity boom on consumption for Ricardian consumers. This effect is initially increasing and reaches a peak 17 quarters after the shock. This increase is slightly higher under the countercyclical fiscal rule due to the higher labor income.

The effect on non-Ricardian consumption is shown in Fig. 5. It is qualitatively similar to that in Fig. 4, but in this case the initial effect is lower under countercyclical fiscal rules due to the initial reduction in government transfers. However, Fig. 5 also shows that this positive effect on consumption dissipates more rapidly under the balanced-budget rule since there are no additional government savings in this case. In other words, this fiscal countercyclicality allows a smoother consumption dynamics for non-Ricardian agents.

In summary, countercyclical fiscal rules allow non-Ricardian consumers to enjoy a smoother effect on consumption which is welfare improving despite the initial decrease in government transfers which is partly compensated by further hours of work. The differential effect on wages allows Ricardian consumers to enjoy fewer working hours

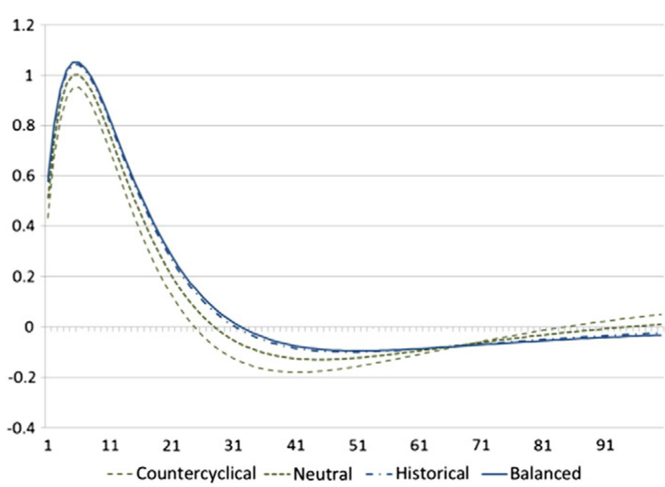

Fig. 1. Effects on Ricardian consumers' employment: percent deviations from steady state under each fiscal policy rule.

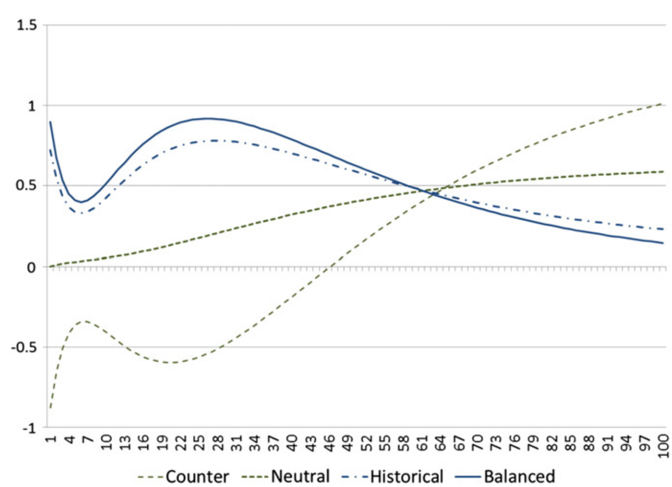

Fig. 3. Effects on government transfers: percent deviations from steady state under each fiscal policy rule.

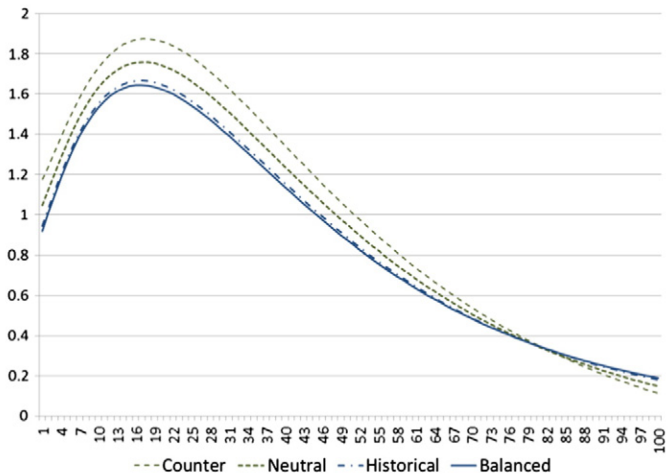

Fig. 4. Effects on Ricardian consumption: percent deviations from steady state under each fiscal policy rule.

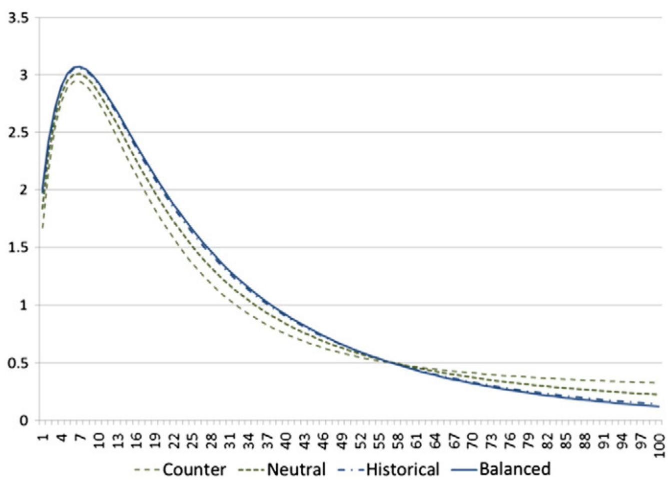

Fig. 5. Effects on non-Ricardian consumption: percent deviations from steady state under each fiscal policy rule.

and higher consumption during the development of the commodity boom under countercyclical rules.[18]

6. Welfare analysis

We now present the welfare evaluation of different cyclicality levels of the fiscal rule, for each type of consumer. We calculate the welfare change, for each cyclicality degree, as the percentage of consumption that makes households indifferent between being in a steady state economy[19] or being in an economy that experiences productivity shocks in the commodity sector taking into account the unconditional distribution of that shock. The size of compensated consumption is calculated then as the proportion π^h such that

$$E_0\sum_{t=0}^{\infty}\left(\beta^t u\left(C_0^h\left(1+\pi^h\right),L_0^h\right)\right)=E_0\sum_{t=0}^{\infty}\left(\beta^t u\left(C_t^h,L_t^h\right)\right)$$

where variables indexed with 0 are those of the steady state. The proportion π^h is calculated from the solution to the equilibrium conditions of the model using a second-order approximation around the steady state, following the methodology by Schmitt-Grohé and Uribe (2004) and Kim et al. (2008).

Table 2
Welfare change, means and standard deviations under different fiscal cyclicality levels.

Fiscal rule cyclicality (φ)	A. Ricardian consumers				
	100*Δπ	Labor supply		Consumption	
		Mean	SD	Mean	SD
−1	−2.42%	0.89	1.26	−1.44	1.28
−0.5	−1.65%	0.61	1.11	−1.00	1.19
0	−0.99%	0.38	1.02	−0.61	1.11
0.5	−0.45%	0.17	0.98	−0.28	1.05
1	0.00%	0.00	1.00	0.00	1.00

Fiscal rule cyclicality (φ)	B. Non-Ricardian consumers					C. Government transfers	
	100*Δπ	Labor supply		Consumption			
		Mean	SD	Mean	SD	Mean	SD
−1	1.72%	−0.90	8.61	1.29	1.00	7.78	4.96
−0.5	1.24%	−0.62	6.39	0.88	0.98	5.38	3.64
0	0.79%	−0.38	4.28	0.53	0.97	3.30	2.42
0.5	0.38%	−0.18	2.34	0.24	0.98	1.51	1.38
1	0.00%	0.00	1.00	0.00	1.00	0.00	1.00

Source: authors' calculations.

We report the results in Table 2. In column 2, we present the difference between the value of π^h calculated for each rule and the one calculated for the balanced-budget case ($\varphi = 1$). In the other columns, we report the mean and standard deviation of labor and consumption (for each type of consumer) as well as those for government transfers. Means are presented as the percentage change relative to the mean of the variable under the balanced-budget rule. Standard deviations are shown as the ratio relative to the standard deviation under balanced budget.

From panel A, we conclude that Ricardian consumers prefer a balanced budget over any other cyclicality degree. In contrast, from panel B, we see that non-Ricardian consumers prefer a countercyclical rule. These results are standard in the literature on fiscal rules (e.g. García et al., 2011, and García-Cicco and Kawamura, 2015). For the Colombian case, we find that moving from a balanced-budget rule to a completely countercyclical rule ($\varphi = -1$) decreases in 2.42 percentage points the Ricardian's consumption-equivalent welfare relative to the steady state. The same movement increases in 1.72 percentage points the non-Ricardian's welfare.[20] These values are relatively small compared to the literature, a fact that is mostly explained by the moderate shock persistence estimated using Colombian data (i.e. $\varrho_e = 0.79$; see Section 4).

[18] Although we focus on welfare effects of different fiscal rules, it is worth mentioning that, in our model, the productivity shock has an initial negative effect on manufacturing output which lasts four quarters only (it is not shown here). This result is opposed to the standard in the literature on Dutch Disease where such negative effect lasts longer. A sensitivity analysis shows this is explained by the fact that our estimates of both the persistence of the productivity shock ($\varrho_e = 0.79$) and the elasticity of manufacturing exports to the real exchange rate ($\xi = 0.32$) are relatively low. For instance, in Ojeda et al. (2014), where $\varrho_e = 0.97$ and $\xi = 0.63$, the negative effect on manufacturing output lasts several years.

[19] The economy's steady state is independent of the cyclicality parameter of the fiscal rule.

[20] We calculate the expected welfare levels for Ricardian agents under different fiscal rules in a model with Ricardian consumers only and in a model where they represent 50% of the consumers in the economy. We find that: a) Ricardian's steady-state welfare is lower in an economy with only this consumer type than in a model where non-Ricardian consumers are present; b) given the specification for fiscal policy used in our model, Ricardian consumers are indifferent among fiscal rules when they are the only agent type in the economy; c) as long as the parameter of fiscal cyclicality is not too low (approximately, $\varphi \geq -0.4$), expected welfare for Ricardian consumers of experimenting shocks under such fiscal rule is higher when there are non-Ricardian consumers.

For Ricardian consumers, as the fiscal rule becomes more procyclical, the mean of labor supply decreases and that of consumption increases. The opposite happens in the case of non-Ricardian consumers. Moreover, fiscal procyclicality reduces labor volatility for both types of consumers but reduces consumption volatility for Ricardian consumers only. It also reduces the mean and the standard deviation of government transfers.

7. Conclusions

We analyze the effects on a small open commodity-rich economy of a temporary shock in the commodity sector productivity under alternative fiscal rules. We build a DSGE model with three productive sectors (one service or non-tradable sector and two tradable ones: a manufacturing sector and a commodity sector) plus government and two types of consumers differentiated by their access to financial markets (Ricardian and non-Ricardian).

We estimate some model parameters by Bayesian methods using Colombian macroeconomic quarterly data for the period 1996–2011. The remaining parameters are fixed or calibrated in line with stylized facts of the Colombian economy or based on the related literature.

We conclude that while Ricardian consumers prefer a balanced budget over any other cyclicality degree, non-Ricardian consumers prefer a countercyclical rule. For Colombia, we find that moving from a balanced-budget rule to a completely countercyclical one decreases in 2.42 percentage points the Ricardian's consumption-equivalent welfare relative to the steady state. The same movement increases in 1.72 percentage points the non-Ricardian's welfare. These values are relatively small compared to the literature, a fact that is mostly explained by the moderate shock persistence estimated using Colombian data.

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